

Jonathan Acosta

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EDUCATION

Columbia University

MS in Mechanical Engineering, GPA: 3.79

2025 GEM Fellow

New York, NY

Dec 2026

University of Florida

BS in Mechanical Engineering

Gainesville, FL

Dec 2024

PROFESSIONAL EXPERIENCE

Ford Motor Company

Dearborn, MI

Robotics Product Intern

June 2026 - Aug 2026

- Designed and programmed an open-perimeter safety system using AutoDesk Inventor and SICK PLC Safety Designer, integrating laser scanners and wireless E-stops across fail-safe fault states to replace physical cell guarding.
- Engineered an EOAT gripper for an AMR manipulator, maintaining <1-inch operational clearance while limiting mechanical deflection during dynamic payload transfers.
- Executed full tear downs and rebuilds of prototype manipulators, applying DFM/DFA modifications that eliminated wire harness pinch points and minimized maintenance cycle times.

Global Lean Manufacturing Strategist Intern

May 2025 - Jul 2025

- Created and standardized manufacturing workflows across 3 production lines, reducing recurring issue-resolution time by 30%.
- Implemented contamination source mapping and OEE/OLE performance metrics to identify production losses and standardize quality and efficiency procedures.
- Developed operator training tools and maintenance-path visualizations to improve execution consistency and line accuracy by 20%.

Tesla

Fremont, CA

Manufacturing Equipment Engineer Intern

Jan 2024 - Jul 2024

- Designed and built sensor mounts to detect datum drift and prevent motor issues, cutting down battery pack scrap by 3 per month.
- Fabricated additively manufactured jigs to discharge batteries for cell deformation testing, at a rate of 8 cells per 30 minutes.
- Eliminated 5 daily unscheduled downtimes by redesigning RFID reader fixtures with weldments instead of fasteners to minimize operational vibrations.
- Saved \$200k annually in scrap by utilizing proximity sensors and fixtures to track container positions and expiration dates.
- Reduced headcount by 1 and saved \$100k annually by applying GD&T to design a high precision locating plate for auto cell pickup.
- Produced a spring-loaded scale to alert associates when purge bucket dropped below 25 lbs., preventing overfill and enhancing safety.

CooperSurgical Inc.

Trumbull, CT

Manufacturing Engineer Intern

May 2023 - Aug 2023

- Led task of cataloging all 522 assets in facility through a third-party system and worked with technicians to create maintenance schedules for each asset.
- Boosted overall line efficiency (OLE) by 7% using SolidWorks and 3D printing for rapid prototyping of manufacturing improvements.
- Increased assembly efficiency by 15% by prototyping and refining tool designs with the validation team.

Formula SAE (Gator Motorsports)

Gainesville, FL

Tractive System Engineer

Jul 2022 - Dec 2023

- Calculated a Final Drive Ratio (FDR) of 2.92 for the EV, by creating a MATLAB lap time simulator and examining which FDR produced fastest time on a given track.
- Designed and assembled a scatter shield from 12GA steel for sprocket and chain, and a motor shield from 18GA aluminum for crash protection and safety.
- Determined desired lengths of half shafts (used to transmit torque) using hand calculations to machine down to wanted size and mark where tripod bearings would be housed.

ACADEMIC/TECHNICAL PROJECTS

Cold Plate Optimization: Conducted thermal analysis of an EV battery cold plate using 3D finite difference modeling on Python and Ansys to evaluate which material and serpentine coolant geometry dissipated the most heat.

Gimbal Pointing System: Developed an autonomous, two-axis, gimbal prototype meant to manipulate solar arrays on LEO satellites to position them to capture maximum solar energy.

Ammonia-Fueled Supersonic Jet: Determined the feasibility of ammonia replacing Jet-A as a fuel source for planes by simulating the combustion process of ammonia in a supersonic jet and comparing the thrust, range, and thrust specific fuel consumption.

SKILLS

Design: SolidWorks, AutoCAD, Fusion 360, ANSYS, Technical Drawings, GD & T, FEA, FMEA, Siemens NX, Autodesk Inventor

Programming: MATLAB, Microsoft Excel, Python, PLC (Allen Bradley Ladder Logic), FANUC (Teach Pendant), SICK PLC

Manufacturing: Manual Mill and Lathe, Bandsaw, Drill Press, Sheetmetal Fabrication, 3D Printing, Lean Six Sigma